

Build in 3D

TODAY YOU'LL MAKE

Something solid you can spin around on screen.

GET READY

- ☐ Laptop
- ☐ Mouse
- ☐ Tinkercad login
- ☐ Pencil

STEPS

- 1** Sign in. Click **Create**, then **3D design**.
- 2** Drag a **Box** onto the workplane. Pull its corners bigger.
- 3** Drag a **Cylinder** in. Switch it to **Hole**. Push it into the box.
- 4** Select both. Click **Group**. Watch the hole appear.
- 5** Hold right-click and spin it around. Look underneath.

IF IT BREAKS

Hole didn't cut? Both shapes have to be selected before you click Group.

Shape is floating? Drag the little black cone on top down to zero.



SCAN ME

YOUR CHALLENGE

Build a house with a door and two windows. Every opening is a **Hole**.

DRAW YOUR HOUSE FIRST



I DID IT!

Date: _____

ONE THING I LEARNED

Make It Yours

TODAY YOU'LL MAKE

A name tag with your own name on it.

GET READY

- ☐ Laptop
- ☐ Mouse
- ☐ Tinkercad login
- ☐ Pencil

STEPS

- 1** Start a new **3D design**.
- 2** Drag in a **Box**. Type its size: **60** by **20** by **3** mm.
- 3** Drag **Text** on top. Type your name. Make it **2** mm tall.
- 4** Add a **Cylinder** hole near one end for a keyring.
- 5** Select everything. Click **Group**.

IF IT BREAKS

Letters sank into the box? Lift the text to 3mm. It sits on top, not inside.

Name too long? Make the box longer, or make the text narrower.



SCAN ME

YOUR CHALLENGE

Make one for everybody in the house. Give each person a different shape.

DRAW THE TAG, FULL SIZE

**I DID IT!**

Date: _____

ONE THING I LEARNED

Measure Twice

TODAY YOU'LL MAKE

A stand built to fit a real object.

GET READY

- ☐ Laptop
- ☐ Mouse
- ☐ Ruler
- ☐ micro:bit
- ☐ Pencil

STEPS

- 1** Measure your micro:bit with a ruler. Write the mm here.
- 2** Start a new design. Build a stand for it to sit in.
- 3** Make the slot **1 mm wider** than you measured.
- 4** Set the slot shape to **Hole**, select all, then **Group**.
- 5** Spin it round. Check the slot goes all the way through.

IF IT BREAKS

Came out tiny? Tinkercad counts in mm. 5 cm is **50**.

Slot off to one side? Use **Align** to centre it, then nudge with the arrow keys.



SCAN ME

YOUR CHALLENGE

Design a stand for something else — a phone, a book, the dog's brush. Measure first.

WRITE YOUR MEASUREMENTS IN MM

**I DID IT!**

Date: _____

ONE THING I LEARNED

Send It to the Printer

TODAY YOU'LL MAKE

A design on screen turned into a real object.

GET READY

- ☐ Laptop
- ☐ Mouse
- ☐ 3D printer
- ☐ Slicer open
- ☐ Pencil

STEPS

- 1 Open your name tag from Week 17.
- 2 Click **Export**, then **.STL**.
- 3 Open that file in the slicer. Click **Slice**.
- 4 Read the time it gives you. Write it down here.
- 5 Send it to the printer. Watch the whole first layer go down.

IF IT BREAKS

Export greyed out? Click your design once to select it first.

Slicer says it's off the bed? Click **Centre**, then slice again.



SCAN ME

YOUR CHALLENGE

Guess the print time before the slicer tells you. How close were you?

MY GUESS / THE REAL TIME

**I DID IT!**

Date: _____

ONE THING I LEARNED

Print It and Fix It

TODAY YOU'LL MAKE

A second version that's better than the first.

GET READY

- ☐ Laptop
- ☐ Your printed part
- ☐ 3D printer
- ☐ Ruler
- ☐ Pencil

STEPS

- 1** Hold your print. Write down **three** things you'd change.
- 2** Open the design again. Fix the worst one.
- 3** Make anything thin at least **2 mm** thick.
- 4** Export it and slice it again.
- 5** Print it. Stand version 1 and version 2 side by side.

IF IT BREAKS

It snapped? Thin parts break. Go thicker, or lay it flat on the bed.

Letters are fuzzy? Raise the text a little and print it slower.



SCAN ME

YOUR CHALLENGE

Show both to someone and let them guess which came first. Ask them why.

DRAW V1 AND V2 SIDE BY SIDE

**I DID IT!**

Date: _____

ONE THING I LEARNED
